

Focalization in situ and the absence of covert focus movement in Brazilian Portuguese

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Abstract: Brazilian Portuguese allows focalization in situ, in the sense that any sub-constituent of the sentence can be focalized in the canonical order, being solely marked in the prosody. The paper addresses the question of whether foci in situ must undergo covert movement to a dedicated focus position, in order to assess the validity of Rizzi's (1997) Focus Criterion in this language. First, it is argued that postverbal focalization is done truly in situ in overt syntax, without involving movement (either overtly or covertly) to Belletti's (2004) dedicated focus position (FocP) in the periphery of vP. Then, based on phenomena involving negative concord and contrastive topicalization, it is argued that focalization in situ does not involve covert movement to Rizzi's (1997) dedicated focus position (FocP) in the periphery of CP. The paper therefore maintains that the Focus Criterion is not operative in Brazilian Portuguese.

Keywords: *focalization; focus criterion; covert movement; negative concord; contrastive topicalization.*

1. Introduction

Brazilian Portuguese allows focalization in situ, in the sense that the focalization of any sub-constituent of the clause is possible in the canonical word order, which is standardly assumed to be subject–verb–object (SVO), with the direct object preceding the indirect object in ditransitive constructions (see e.g. Scher 1996; Armelin 2011; and Cépeda & Cyrino 2020). The canonical order is evidenced by the *what happened* test, which elicits an all-new sentence (i.e., broad focus), as in (1).

- (1) A: O que aconteceu?
 ‘What happened?’
 B: O João deu um livro pra Maria.
 the John gave a book to-the Mary
 ‘John gave a book to Mary.’

As is shown in the dialogues in (2)–(4) below, the S–V–DO–IO order allows the focalization of the subject, the direct object, and the indirect object, respectively, with the relevant focalized elements all being marked solely in the prosody.¹

¹ The exhaustive particle *só* ‘only’ is used here to facilitate focus interpretation. While overt fronting of plain focus is judged marginal in Brazilian Portuguese, overt fronting of exhaustive focus is usually acceptable; therefore, in order to assess the possibility of covert movement, I made the choice of using examples that would be amenable to overt movement, for a fair comparison.

- (2) A: Quem deu um livro pra Maria?
 ‘Who gave a book to Mary?’
 B: Só o João_F deu um livro pra Maria.
 only the John gave a book to-the Mary
 ‘Only John_F gave a book to Mary.’
- (3) A: O que o João deu pra Maria?
 ‘What did John give to Mary?’
 B: O João deu só um livro_F pra Maria.
 the John gave only a book to-the Mary
 ‘John gave only a book_F to Mary.’
- (4) A: Pra quem o João deu um livro?
 ‘Who did John give a book to?’
 B: O João deu um livro só pra Maria_F.
 the John gave a book only to-the Mary
 ‘John gave a book only to Mary_F.’

In addition to focalization in situ, Brazilian Portuguese also allows focus fronting. Sentences such as (3B) and (4B) above often have left-peripheral counterparts, as in (5a) and (5b), respectively.

- (5) a. *Só um livro_F o João deu pra Maria.*
 only a book the John gave to-the Mary
 ‘*Only a book_F John gave to Mary.*’
- b. *Só pra Maria_F o João deu um livro.*
 only to-the Mary the John gave a book
 ‘*Only to Mary_F John gave a book.*’

In the literature on Brazilian Portuguese, several authors adopt the so-called *cartographic* approach to information structure, following the tradition of Rizzi (1997, 2004, 2006, 2010, 2015, 2018) and Belletti (2001, 2004), and analyze focalization in this language as involving movement of the relevant element to a dedicated focus position (i.e., Spec,FocP) in the clausal spine (see e.g. Miotto 2001, 2003; Quarezemin 2009; Armelin 2011; Kato 2013, 2020; Guesser & Quarezemin 2013; Guesser 2015; Quarezemin & Tescari Neto 2015; Cépeda & Cyrino 2020, among others). Postverbal focalization is analyzed as involving movement to a focus position in the periphery of vP, as in (6), whereas focus fronting is analyzed as involving movement to a position in the periphery of CP, as in (7).

- (6) a. $[_{TP} \text{ o João deu } [_{FocP} \text{ só um livro}_F [_{VP} [_{VP} \text{ t}_F \text{ pra Maria}]]]]$ cf. (3B)
- b. $[_{TP} \text{ o João deu } [_{TopP} \text{ um livro}_i [_{FocP} \text{ só pra Maria}_F [_{VP} [_{VP} \text{ t}_i \text{ t}_F]]]]]$ cf. (4B)

- (7) a. $[_{\text{FocP}} \text{Só um livro}_F [_{\text{TP}} \text{o João deu } t_F \text{ pra Maria}]]$ cf. (5a)
 b. $[_{\text{FocP}} \text{Só pra Maria}_F [_{\text{TP}} \text{o João deu um livro } t_F]]$ cf. (5b)

In the cartographic approach, the movement of a focus to Spec,FocP takes place in order to comply with the *Focus Criterion* (Rizzi 1997), which is the interface requirement stated in (8). The role of the Focus Criterion is to ensure the creation of a focus-presupposition articulation of the clause, as structurally represented in (9).

(8) *Focus Criterion:*

A constituent endowed with a focus feature must end up in a Spec-head configuration with Foc^0 .

(Rizzi 1997: 287, adapted)

- (9) $[_{\text{FocP}} \text{Focus}_i [_{\text{Foc}'} \text{Foc}^0 [_{\text{Presupposition}} t_i]]]$

(Rizzi 1997: 287, adapted)

Like in Brazilian Portuguese, the canonical order in European Portuguese is also SVO. Unlike in Brazilian Portuguese, focus in European Portuguese cannot be freely assigned to any constituent in the canonical order, but must instead appear in final position (see e.g. Kato, Martins & Nunes 2022 and references therein). As is shown in (10), for instance, the SVO order does not allow focalization of the subject (cf. (2B) above), whereas the VOS order, which is ungrammatical in Brazilian Portuguese, is acceptable in this scenario.

(10) A: Quem é que partiu a janela? (EP)

‘Who broke the window?’

B1: #O Paulo partiu a janela.

the Paulo broke the window

B2: Partiu a janela o Paulo.

broke the window the Paulo

‘Paulo broke the window.’

(Costa 2000: 197)

Costa (2000) analyses focalization in European Portuguese with the focus being licensed in situ (in the spirit of Rooth 1985), with non-focal elements moving overtly to positions before the focus, ensuring that the focus appears in final position. In particular, the author raises the question of whether foci in situ in European Portuguese must undergo movement to the left periphery of the clause at LF and argues that the postulation of covert focus movement in this language is both conceptually and empirically inadequate.

Comparing the analysis of focalization in Brazilian Portuguese in the cartographic tradition (see references above) with Costa’s (2000) analysis of focalization in European Portuguese, we may then wonder whether there is a parametric point of variation between the two varieties of the language, in that Rizzi’s (1997) Focus Criterion is operative in the former but not in the latter.

In this paper, I will assess the validity of the Focus Criterion in Brazilian Portuguese, by addressing the question of whether focalization must involve movement to a dedicated focus position in the clausal spine. In particular, I will address the question of whether postverbal focalization in situ in Brazilian Portuguese, as seen in sentences such as (3B) and (4B) above, involves covert focus movement to the left periphery of the clause, in order to comply with the Focus Criterion at LF, as in the putative structures in (11).

- (11) a. *Overt:* [_{TP} O João deu só *um livro*_F pra Maria] cf. (3B)
 LF: [_{FocP} só *um livro*_F [_{TP} o João deu *t_F* pra Maria]]
- b. *Overt:* [_{TP} O João deu um livro só *pra Maria*_F] cf. (4B)
 LF: [_{FocP} só *pra Maria*_F [_{TP} o João deu um livro *t_F*]]

The paper is organized as follows. In Section 2, I will briefly review Costa’s arguments that focalization in situ in European Portuguese does not involve covert focus movement. In Section 3, I will first briefly review Lacerda’s (2024a) arguments that postverbal focalization in Brazilian Portuguese is truly in situ in overt syntax and does not involve movement to a focus position in the periphery of vP (cf. (6) above); then, I will argue that the vP-peripheral analysis of postverbal focalization faces problems in the creation of a proper focus-presupposition articulation at LF (cf. (9) above). That discussion will lead us to Section 4, where I will address the paper’s main question of whether foci in Brazilian Portuguese must move to the left periphery of the clause at LF (cf. (11) above). Based on novel evidence from two independent empirical domains — namely, negative concord and

contrastive topicalization —, I will argue that the postulation of covert focus movement is inadequate in Brazilian Portuguese. Finally, Section 5 ends the paper with the conclusion that Rizzi's (1997) Focus Criterion is not operative in Brazilian Portuguese, there being no parametric variation with respect to European Portuguese when it comes to covert focus movement.

2. European Portuguese

Costa (2000) raised the question of whether foci in situ in European Portuguese must undergo movement to the left periphery of the clause at LF — the precise question I will address with respect to Brazilian Portuguese in this paper. For the author, the postulation of covert focus movement is conceptually undesired in European Portuguese, because it uneconomically duplicates the process of identifying foci in this language; he defends that a prosodic algorithm read off overt syntax is sufficient for the correct and transparent identification of the focus.

Based on the rearrangement of the word order in overt syntax, focus interpretation can be assigned to “the prosodically most prominent constituent plus everything it c-commands” (p. 203; see the work cited for the precise implementation of the algorithm). As we saw in the Introduction, focalization of the subject is not possible in the canonical SVO order, but is possible in the VOS order, as in (12B) below. By the same algorithm, VSO order is employed when both the subject and the object are interpreted as focus (and the verb as old information), the subject being the prosodically most prominent constituent

and c-commanding the object, as in (13B). Covert focus movement would dismantle the correct c-command relations created in overt syntax for the identification of the focus, which Costa (2000) views as an unreasonable operation conceptually (p. 209).

(12) A: Quem é que partiu a janela? (EP)

‘Who broke the window?’

B: Partiu a janela o Paulo. VOS

broke the window the Paulo

‘Paulo broke the window.’

(Costa 2000: 197)

(13) A: Ninguém partiu nada. (EP)

‘No one broke anything.’

B: Partiu o Paulo a janela. VSO

broke the Paulo the window

‘Paulo broke the window.’

(Costa 2000: 197)

Costa (2000) further argues that the postulation of covert focus movement in European Portuguese is empirically inadequate. By pointing out that the subject is the prosodically most prominent element in both VOS and VSO orders, the author entertains two kinds of LF representations: one where only the most prosodically salient element moves to a focus

position and one where all elements interpreted as focus move (which would in turn be closer to a cartographic analysis). In the first option, only the subject would covertly move to a focus position in both (12B) and (13B) above, as represented in (14) below, thus creating indistinguishable LF representations for sentences that have different interpretations and failing to capture the fact that the object is new information in the VSO order but old information in the VOS order.

- (14) LF: [_{FocP} O Paulo [_{TP} partiu a janela]] cf. (12B) and (13B)
(Costa 2000: 208–209, adapted)

A second possibility is the one where all the elements interpreted as focus move to a focus position at LF. The structure in (14) above would thus represent the LF structure of only the VOS order in (12B) and the LF structure of the VSO order in (13B) would be the one in (15).

- (15) LF: [_{FocP} [a janela]_i [o Paulo]_j [_{TP} partiu t_j t_i]] cf. (13B)
(Costa 2000: 211, adapted)

The structure above, though unambiguous, would still be problematic, Costa (2000) claims. Assuming with Pesetsky (1982) that crossing paths are not allowed, the author argues that the LF representation of sentence (13B) must be the one in (15) above and cannot be the one in (16) below.

(16) LF: *_[FocP] [o Paulo]_j [a janela]_i [TP partiu t_j t_i]] cf. (13B)

(Costa 2000: 211, adapted)

In the absence of crossing paths, the assumption of covert focus movement as in (15) above makes the prediction that the scope relation between the subject and the object at LF must necessarily be reversed in VSO structures, Costa (2000) argues. Contrary to the prediction, though, sentence (17) allows both scope possibilities, whereas sentence (18) only allows wide scope of the subject.

(17) Leram dois rapazes um livro. (S>O, O>S) (EP)

read two boys a book

‘Two boys read a book.’

(Costa 2000: 211)

(18) Leu cada professor um exame. (S>O, *O>S) (EP)

read each professor an exam

‘Each professor read an exam.’

(Costa 2000: 211)

Based on the above and other arguments, Costa (2000) concludes that there is no covert focus movement in European Portuguese and that foci in situ are interpreted in sentence-internal positions at LF, in the spirit of Rooth (1985).

Due to crucial differences between European Portuguese and Brazilian Portuguese, Costa's (2000) argumentation cannot be straightforwardly carried over to Brazilian Portuguese, making it unviable to establish a direct comparison between the two varieties of the language in light of his arguments. The bulk of the author's argumentation depends on the availability of VOS and VSO orders in European Portuguese (cf. (12) and (13) above), which are not possible in Brazilian Portuguese, as respectively shown in (19) and (20) below (adapted from Kato 2000).²

(19) A: Quem é que assinou uma carta? (BP)

‘Who signed a letter?’

B: *Assinou uma carta o chefe do departamento. *VOS

signed a letter the head of-the department

‘The head of the department signed a letter.’

(Kato 2000: 97, adapted)

(20) A: Ninguém assinou nada. (BP)

‘No one signed anything.’

² See Lacerda (2024b) for a deduction of the absence of VSO and VOS orders in Brazilian Portuguese.

*VSO

signed the John a letter

‘John signed a letter.’

(Kato 2000: 101, adapted)

Costa's (2000) prosodic algorithm for the identification of the focus in European Portuguese is also not applicable to Brazilian Portuguese. As was shown by the examples (2)–(4) in the Introduction, focus stress may fall upon any constituent in the canonical order in Brazilian Portuguese, which is not the case in European Portuguese.

Considering the crucial differences in the syntax and stress patterns associated with focalization in the two varieties of Portuguese, we may reasonably wonder if these differences lead to a parametric variation in the licensing of focus at LF, in that covert focus movement does not take place in European Portuguese but does so in Brazilian Portuguese. At the same time, we are forced to envision different testing grounds for the availability of covert focus movement in Brazilian Portuguese. As I will argue in the following sections, despite its differences with respect to the European variety, Brazilian Portuguese behaves just like European Portuguese in not displaying covert focus movement in constructions with focalization *in situ*.

3. Focalization in situ

In order to justify addressing the question of whether foci in situ in Brazilian Portuguese must move to the left periphery of the clause at LF to comply with the Focus Criterion, I must first rule out the possibility that the Focus Criterion may be satisfied in the postverbal area through “low focus movement” to Spec,FocP in the periphery of vP.

3.1. *The absence of “low focus movement”*

In analyzing postverbal subjects in Italian, Belletti (2001, 2004) proposes that the periphery of vP includes topic- and focus-designated projections, similarly to Rizzi’s (1997) analysis of the left periphery of the sentence, as represented in (21). When a postverbal subject in Italian has focus interpretation, as in (22B), Belletti (2001, 2004) proposes that it is licensed in the specifier of a focus projection in the periphery of vP, as represented in (23).

(21) [IP [TopP [FocP [TopP [vP]]]]]

(Belletti 2004: 25)

(22) A: Chi ha parlato?

‘Who spoke?’

B: Ha parlato Gianni.

has spoken Gianni

‘Gianni spoke.’

(Belletti 2004: 21)

(23) [TP Ha parlato [_{FocP} *Gianni*_F [_{VP} t_F [_{VP}]]]]

Several works within the cartographic tradition have assumed the existence of topic and focus projections in the periphery of vP in Brazilian Portuguese (see e.g. Miotto 2003; Quarezemin 2009; Armelin 2011; Kato 2013, 2020; Quarezemin & Tescari Neto 2015; Cépeda & Cyrino 2020), especially in analyzing ditransitive and *wh*-in situ constructions.³ For instance, Cépeda & Cyrino (2020) propose that sentence (24B) has the structure in (25), while Kato (2013) proposes that sentence (26a) has the structure in (26b).

(24) A: O que a Olga deu para o Mário?

‘What did Olga give to Mario?’

B: A Olga deu para o Mário uma maçã.

the Olga gave to the Mario an apple

‘Olga gave an apple to Mario.’

(Cépeda & Cyrino 2020: 110)

(25) [TP A Olga_{subj} deu [_{TopP} para o Mario_{top} [_{FocP} uma maçã_{foc} [_{VP} t_{subj} [_{VP} t_{foc} [_{V'} t_{top}]]]]]]

(Cépeda & Cyrino 2020: 112)

³ For an analysis of *wh*-in situ that resorts to low focus movement in other languages, see Bonan (2021).

- (26) a. João tinha restituído para a Maria que livro?
 John had returned to the Mary which book
 ‘Which book had John returned to Mary?’
- b. João tinha restituído [_{TopP} para a Maria_i [_{FocP} que livro_w [_{vP} [_{VP} t_w t_i]]]]
- (Kato 2013: 183, adapted)

Contrary to the above works, Lacerda (2024a) presents a number of arguments in favor of analyzing postverbal focalization in Brazilian Portuguese as being truly in situ in overt syntax, without involving “low focus movement” to the periphery of vP, thus not being subject to the Focus Criterion in overt syntax. I will present here one argument regarding ditransitive constructions and one argument regarding *wh*-in situ and will refer the reader to the work cited for a much lengthier discussion.

Regarding ditransitive constructions in Brazilian Portuguese, Quarezemin (2009) notes that focalization of the indirect object in the non-canonical IO_F–DO order is marginal when the direct object is indefinite, as in (27B). Lacerda (2024a) notes, however, that focalization of the indirect object in the IO_F–DO order is perfectly acceptable when the direct object is definite, as in (28B).

- (27) A: Pra quem o João deu um presente?
 ‘Who did John give a gift to?’
- B: ??O João deu só pra Maria_F um presente.
 the John gave only to-the Mary a gift

‘John gave a gift *only to Mary_F*.’

(Quarezemin 2009: 117, adapted)

(28) A: Pra quem o João deu o Programa Minimalista?

‘Who did John give The Minimalist Program to?’

B: O João deu só *pra Maria_F* o Programa Minimalista.

the John gave only to-the Mary the Program Minimalist

‘John gave The Minimalist Program *only to Mary_F*.’

(Lacerda 2024a: 217)

Lacerda (2024a) takes the contrast between (27B) and (28B) to indicate that the non-canonical IO_F–DO order (with the indirect object focalized) in Brazilian Portuguese is not derived by leftward low focus movement of the indirect object past the direct object, which should in fact be immaterial to the definiteness of the direct object, but rather by right dislocation of the direct object, which is a kind of topicalization of discourse-given material. The author shows, based on a number of examples, that direct objects in the IO_F–DO order exhibit the same semantic restrictions (related to definiteness and referentiality) that subjects do in subject right dislocation. As is illustrated in the contrast between (29) and (30), right dislocation of subjects in Brazilian Portuguese is acceptable with a definite subject, but marginal with an indefinite one (and completely ruled out with quantified subjects; see Lacerda 2024a).

(29) A: Quantos livros o João leu?
‘How many books did John read?’

B1: O João leu três livros.
the John read three books

B2: Leu três livros, o João.
read three books the John
‘John read three books’

(Lacerda 2024a: 219)

(30) A: Qual livro um aluno leu?
‘Which book did a student read?’

B1: Um aluno leu o *Dom Casmurro*.
a student read the Dom Casmurro.

B2: ??Leu o *Dom Casmurro*, um aluno.
read the Dom Casmurro a student
‘A student read *Dom Casmurro*.’

(Lacerda 2024a: 220)

Regarding *wh*-questions, Lacerda (2024a) notes a contrast between *wh*-fronting and *wh*-in situ. In constructions with *wh*-fronting, both the pied-piping structure in (31a) and the sub-extraction structure in (31b) are acceptable; possible structures for (31a) and (31b) are given in (32a) and (32b), respectively.

- (31) a. Um livro *de que autor* o João resenhou (ontem)?
 a book of which author the John reviewed (yesterday)
- b. *De que autor* o João resenhou um livro (ontem)?
 of which author the John reviewed a book (yesterday)
- ‘John reviewed a book *by which author* (yesterday)?’

(Lacerda 2024a: 232)

- (32) a. [_{FocP} [Um livro *de que autor*]_i [_{TP} o João resenhou [_{VP} [_{VP} *t_i*]]]]
- b. [_{FocP} [*De que autor*]_i [_{TP} o João resenhou [_{VP} [_{VP} [um livro *t_i*]]]]]]

(Lacerda 2024a: 233)

When the relevant *wh*-expression appears in the postverbal area, on the other hand, the situation is different. While sentence (33a) is acceptable, sentence (33b) is ruled out. If (33a) had a structure along the lines of (34a), involving low focus movement, sentence (33b) should also be acceptable, contrary to fact, given the availability of the structure in (34b), which would be the low counterpart of the grammatical (32b). The gap in the paradigm is thus left unexplained under the assumption of low focus movement.

- (33) a. O João resenhou um livro *de que autor* (ontem)?
 the John reviewed a book of which author (yesterday)
- b. *O João resenhou *de que autor* um livro (ontem)?

the John reviewed of which author a book (yesterday)

‘John reviewed a book by *which author* (yesterday)?’

(Lacerda 2024a: 232)

- (34) a. [TP O João resenhou [_{FocP} [um livro *de que autor*]_i [_{vP} [_{VP} t_i]]]]
b. *[TP O João resenhou [_{FocP} [*de que autor*]_i [_{vP} [_{VP} [um livro t_i]]]]]

(Lacerda 2024a: 233)

Based on the above, Lacerda (2024a) concludes that there is no low focus movement to the periphery of vP in Brazilian Portuguese, contra the literature following Belletti (2001, 2004), and that postverbal focalization in this language is realized truly in situ in overt syntax (see the work cited for additional arguments and further discussion).

3.2. “*Low focus movement*” and the presupposition problem

In this section, I will extend Lacerda’s (2024a) argumentation against low focus movement in Brazilian Portuguese. I will argue that this kind of structure fails to establish a proper focus-presupposition articulation of the clause in the postverbal area. In particular, I will argue that a focus (putatively) sitting at the edge of vP, much like a truly in situ one, does not partake in a bipartite focus-presupposition articulation of the clause, as would be mandated by the Focus Criterion (cf. (9) above).

As was discussed in the previous section, the cartographic approach assumes that both left-peripheral and postverbal focalization involve the creation of a focus-presupposition bipartition of the clause, by means of the projection of a FocP above TP or above vP, as respectively schematized in (35) (see Rizzi 1997, 2004, 2006, 2010, 2015, 2018; Belletti 2001, 2004). Observe that, crucially, the distinct structures in (35a) and (35b) tacitly implicate that a left-peripheral focus and its postverbal counterpart are associated with distinct presuppositions, represented by TP and vP, respectively, where one includes tense and aspect specification (and other TP-related information and occasional adjuncts) and the other does not.

- (35) a. [FocP *focus* [TP *presupposition*]] *left-peripheral focalization*
 b. [TP [FocP *focus* [vP *presupposition*]] *postverbal focalization*

The interpretive distinction predicted to be possible by the structures in (35) is, however, not attested. The presupposition associated with the left-peripheral focus in (36B1), for example, is semantically identical to that associated with its postverbal counterpart in (36B2). In discourse, focalization in both declarative sentences (36B1) and (36B2) is a strategy to resolve the exact same gap (i.e., to identify the variable) posed by the question in (36A). In other words, both (36B1) and (36B2) are associated with the exact same presupposition stated in (37), crucially including tense and aspect information, which is not part of the vP (cf. (35b) above).

- (36) A: O que você pediu?
 ‘What did you order?’
 B1: *Só lasanha_F* eu pedi.
 only lasagna I ordered
 ‘*Only lasagna_F* I ordered.’
 B2: Eu pedi *só lasanha_F*.
 I ordered only lasagna
 ‘I ordered *only lasagna_F*.’

- (37) $\exists x$. I ordered x .

Let us assume, for the sake of the argument, that tense and aspect information could somehow be interpreted as being part of the presupposition associated with a focus in a structure like (35b) above or even that it could be argued not to actually be a necessary part of the presupposition. Still, in order for the vP to qualify as a proper presupposition at LF for a focus sitting at its edge, all the elements overtly moved out of the vP would have to reconstruct back into the vP — more precisely, all the presupposed material would have to be put back under the scope of the focus at LF in order for a covert focus-presupposition configuration to be created in the postverbal area.

Contrary to the above expectations, however, we can readily see that reconstruction of all vP-internal material cannot always be enforced. Suppose that sentence (36B2) above has the LF structure in (38), without reconstruction of the subject and the verb, which have

both moved to the TP domain in overt syntax. The focus *só lasanha* ‘only lasagna’ in (38) is in this case associated with an “empty” vP as its presupposition, an unwelcome result. Thus, (38) cannot be the correct LF structure of (36B2). The subject and the verb must both reconstruct to their vP-internal positions at LF, as in (39), in order to be under the scope of the focus, like in its left-peripheral counterpart in (40).

(38) LF: $*[_{TP} Eu_s \text{ pedi}_v [_{FocP} \text{ só lasanha}_F [_{vP} t_s t_v [_{VP} t_v t_F]]]]$ cf. (36B2)
focus *presupposition*

(39) LF: $[_{TP} \text{ — — } [_{FocP} \text{ só lasanha}_F [_{vP} eu [_{VP} \text{ pedi } t_F]]]]$ cf. (36B2)
focus *presupposition*

(40) LF: $[_{FocP} \text{ Só lasanha}_F [_{TP} eu_s \text{ pedi}_v [_{vP} t_s t_v [_{VP} t_v t_F]]]]$ cf. (36B1)
focus *presupposition*

Though one could presumably argue that verb movement is semantically vacuous and therefore the verb always reconstructs back into the VP, there are still many cases attested where reconstruction of the subject to a position under the postverbal focus is simply not possible.⁴ This is seen in sentences such as (41B) and (42B), where the subject must

⁴ For relevant discussion on the (non-)vacuity of head movement, see e.g. Lechner (2006). Whether the verb in Brazilian Portuguese turns out to be semantically interpreted in the TP or in the VP is immaterial to the present discussion.

distribute over the focus or bind into the focus, respectively. The LF structures in (43) are thus ill-formed.

(41) A: Quantos livros cada aluno leu?
 ‘How many books did each student read?’

B: Cada aluno leu só dois livros_F.
 each student read only two books
 ‘Each student read *only two books*_F.’

(42) A: O que cada aluno apresentou?
 ‘What did each student present?’

B: Cada aluno_i apresentou seu_i melhor artigo_F.
 each student presented their best article
 ‘Each student presented *their best article*_F.’

(43) a. LF: *[TP ___ [FocP só dois livros_F [VP cada aluno [VP leu t_F]]]]
 cf. (41B)

b. LF: *[TP ___ [FocP seu_i melhor artigo_F [VP cada aluno_i [VP apresentou t_F]]]]
 cf. (42B)

Upon closer scrutiny, we see that the low focus movement analysis fails to live up to its promise, namely the creation of a proper focus-presupposition articulation of the clause

in the postverbal area, considering that the vP does not count as a proper presupposition at LF. Assuming that postverbal foci in Brazilian Portuguese are realized truly in situ in overt syntax and cannot involve covert focus movement to a low focus position in the periphery of vP for the reasons mentioned above, we must now raise the question of whether postverbal focalization in situ must comply with Rizzi's (1997) Focus Criterion at LF, through covert movement to a focus position in the left periphery of the clause.

4. Covert focus movement

The idea that focus movement can take place covertly goes back to Chomsky's (1976) classic discussion of weak crossover, as in (44). As (44a) shows, movement of a *wh*-expression across a coindexed possessive pronoun in English gives rise to a weak crossover effect. The contrast between (44b) and (44c) can then be accounted for if (44b) is assumed to involve covert movement of the focalized object, as in (45), as to equate the LF structure of (44a) in the relevant respects.⁵

- (44) a. *Who_i does his_i mother love t_i?
 b. *His_i mother loves JOHN_i.
 c. His_i mother loves John_i.

⁵ But see Rooth (1985) for an account of Chomsky's weak crossover effects without appeal to covert focus movement.

- (45) LF: [*JOHN*_i [his_i mother loves t_i]] cf. (44b)

In Brazilian Portuguese, the counterparts to (44a) and (44b) above are also ungrammatical, as shown in (46a) and (46b), respectively; however, unlike (44c) in English, (46c) is also ungrammatical. Since there is no contrast between (46b) and (46c), the paradigm is silent about the possibility of covert focus movement taking place in (46b).

- (46) a. **Quem*_i a mãe dele_i ama t_i?
 who the mother of-him loves
 ‘*Who_i does his_i mother love?’
- b. **A* mãe dele_i ama *O* *JOÃO*_i.
 the mother of-him loves the John
 ‘*His_i mother loves *JOHN*_i.’
- c. **A* mãe dele_i ama o João_i.
 the mother of-him loves the John
 ‘His_i mother loves John_i.’

It might be the case that sentences (46b) and (46c) above are judged unacceptable due to the fact that the possessive pronoun precedes its antecedent, which is generally bad in Brazilian Portuguese (when reconstruction is not an option). When the antecedent of the

pronoun is controlled for, we can attempt to create the relevant examples using epithets, as in (47) below.⁶ Now, interestingly, both sentences (47B1) and (47B2) are acceptable.

(47) A: E o João_i?

‘What about John_i?’

B1: A mãe dele_i ama o idiota_i.

the mother of-him loves the idiot

‘His_i mother loves the idiot_i.’

B2: A mãe dele_i ama *SÓ O IDIOTA_i* (e mais ninguém).

the mother of-him loves only the idiot (and more nobody)

‘His_i mother loves *ONLY THE IDIOT_i* (and nobody else).’

The acceptability of (47B2) indicates that there is no weak crossover effect at stake. Crucially, (47B2) contrasts with the unacceptable (48B), where the focus moves overtly and expectedly leads to a weak crossover effect. These data therefore suggest that covert focus movement does not take place in (47B2).

(48) A: E o João_i?

‘What about John_i?’

B: **SÓ O IDIOTA_i* a mãe dele_i ama t_i (e mais ninguém)

⁶ I thank Jairo Nunes (p. c.) for his suggestion of using epithets in this test.

only the idiot the mother of-him loves (and more nobody)

‘*ONLY THE IDIOT*_i his_i mother loves (and nobody else).’

I will now probe deeper into the hypothesis that covert focus movement to the left periphery of the sentence is not needed for the proper licensing of foci in Brazilian Portuguese, in order to assess the validity of Rizzi’s (1997) Focus Criterion in this language. The investigation of two independent empirical domains — namely, negative concord and contrastive topicalization — will corroborate this hypothesis.

4.1. *Negative concord*

When proposing the Focus Criterion, Rizzi (1997) pointed out its similarity with the Neg Criterion, that is, the requirement that a negative operator be interpreted in the specifier of a negative phrase (see e.g. Haegeman & Zanuttini 1991; Haegeman 1995):

“we will assume that a constituent endowed with topic or focus features must end up in a Spec/head configuration with Top or Foc, respectively; in other words, there are Topic and Focus Criteria, reminiscent of the Wh and Neg Criteria” (Rizzi 1997: 287)

As is usually assumed, negative concord items must take scope from the position of the sentential negation (see e.g. Zeijlstra 2004 and Giannakidou 2006 for relevant discussion); in other words, negative concord items are subject to the Neg Criterion. That

being so, when a negative concord item is focalized, it should in principle simultaneously satisfy both the Neg Criterion and the Focus Criterion, which leads to a conundrum.

In the cartographic literature, criteria are associated with the so-called “criterial freezing”, that is, the restriction that criteria cannot be satisfied *en passant*. As a consequence, a given element cannot move from one criterial position to another (see e.g. Rizzi 2006, 2010, 2015, 2018 for extensive discussion and possible deductions of freezing effects). Therefore, if both the Neg Criterion and the Focus Criterion are assumed to hold in a given language, a focalized negative concord item cannot comply with both at the same time. Violation of either would then lead to ungrammaticality.

Brazilian Portuguese is a negative concord language, as is evidenced by the fact that sentence (49B) involves a single interpretable negative operator, that is, it is not interpreted as a double negative. Moreover, sentence (49B) shows that the focalization of a negative concord item does not lead to ungrammaticality. In light of this, I will in this section argue that the focalization of negative concord items in Brazilian Portuguese can be accounted for if we keep the Neg Criterion and dispense with the Focus Criterion in this language.

(49) A: Quantos livros você leu?

‘How many books did you read?’

B: Eu não li *nenhum livro*_F.

I not read no book

‘I didn’t read *any book*_F.’

Let us first make the case that negative concord items in Brazilian Portuguese take scope from Spec,NegP, in compliance with the Neg Criterion. Putting aside modal flavors and restricting our discussion to a deontic reading of the modal verb *pode* ‘can’, we observe that sentence (50) is scopally ambiguous, in that either the modal verb can take scope over the quantified direct object *dois livros* ‘two books’ or the direct object can take scope over the modal verb. The first option gives rise to a reading where John is allowed to read a certain number of books, the cardinality two being included; the second option gives rise to a reading where there are two (specific) books which John is allowed to read.

(50) O João pode ler dois livros.

the John can read two books

‘John can read two books.’

pode > *dois livros*; *dois livros* > *pode*

In sentence (51), there is a negative concord item, namely, *nenhum livro* ‘no book’. As opposed to sentence (50) above, sentence (51) has only one interpretation, as far as the scope interaction between the modal verb and the negative operator is concerned. The reading where *pode* ‘can’ takes scope over *nenhum livro* (i.e., John is allowed to read no books at all) is available, whereas the reading where *nenhum livro* takes scope over *pode* (i.e., no book is such that John can read it) is unavailable.⁷

⁷ There being only one interpretable negative operator in negative concord structures, the reading “no book is such that John can *not* read it” is also ruled out, as a reviewer correctly points out.

(51) O João pode não ler nenhum livro.

the John can not read no book

‘John can not read any book.’

pode > *nenhum livro*; **nenhum livro* > *pode*

Given that the negation in (51) is part of the embedded infinitival clause and as such is structurally below the matrix modal verb, *nenhum livro* cannot take scope over *pode*. In compliance with the Neg Criterion, *nenhum livro* must take scope from the position of the overt negation (i.e. Spec,NegP), below *pode*.

Taking the above restriction regarding negative concord to our advantage, we can assess the possibility of covert focus movement in Brazilian Portuguese by analyzing the scope behavior of a focalized negative concord item and seeing whether it is interpreted at LF in the position of the negation, in compliance with the Neg Criterion, or in the left periphery of the sentence, in compliance with the Focus Criterion.

Let us look at the example in (52B), where the direct object is a negative concord item and is focalized in the context of the question in (52A). What is at stake in this dialogue is how many toys the children received and crucially the answer in (52B) only has the surface scope interpretation, whereby there are two children (x and y) which are such that they did not receive any toy.

(52) A: Todas as crianças ganharam *pelo menos um brinquedo*?

‘Did all the children receive *at least one toy*?’

B: Duas crianças não ganharam *nenhum brinquedo*_F.

two children not received no toy

‘Two children did not receive *any toy*_F.’

duas crianças > *nenhum brinquedo*; **nenhum brinquedo* > *duas crianças*

The postulation of covert focus movement of the direct object *nenhum brinquedo* ‘no toy’ in (52B) creates an LF structure along the lines of (53).⁸ Though (53) complies with the Focus Criterion, it incorrectly generates an inverse scope interpretation, whereby no toy was such that two children received it, which is unavailable for (52B) in this scenario, and fails to derive the correct surface scope reading.

(53) LF: [_{FoCP} *nenhum brinquedo*_F [_{TP} duas crianças [_{NegP} t_F não [_{TP} ganharam t_F]]]]

cf. (52B)

Moreover, the purported covert focus movement depicted in (53) violates the Neg Criterion, which requires that *nenhum brinquedo* remain in Spec,NegP at LF in order for its negative quantificational force to be licensed (bearing in mind the criterial freezing). Crucially, note that sentence (52B) is grammatical despite the fact that it cannot respect

⁸ I take NegP to be part of a split IP in Brazilian Portuguese. For concreteness’s sake, I assume that the sentential negation is checked off by the negative quantifier at LF (see e.g. Zeijlstra 2004 and Giannakidou 2006 for relevant discussion).

both the Neg Criterion and the (purported) Focus Criterion at the same time. If we dispensed with the Neg Criterion, we would lose the well-known generalization illustrated in (51) above that negative concord items must take scope from the position of the overt negation. On the other hand, if we dispense with the Focus Criterion, we can assign to sentence (52B) the LF structure represented in (54) below and correctly capture the fact that *nenhum brinquedo* ‘no toy’ takes scope under *duas crianças* ‘two children’.

- (54) LF: [_{TP} duas crianças [_{NegP} *nenhum brinquedo*_F ~~não~~ [_{TP} ganharam t_F]]]
cf. (52B)

In sum, when the Neg Criterion and the Focus Criterion contradictorily “compete” for the licensing of a focalized negative concord item, the data show that the former requirement is valid, whereas the latter can (in fact, must) be dispensed with in Brazilian Portuguese.

Indeed, we find additional evidence for that claim in the interaction of focalized negative concord items and modal verbs (cf. (51) above). Let us look at the contrast in interpretation between sentences (55a) and (55b). In (55a), the matrix clause is negated (i.e., negation is above the modal verb), whereas in (55b) the embedded clause is negated (i.e., negation is below the modal verb). Sentence (55a) thus has the interpretation that John was supposed to give all the books to Mary and he did so (i.e., no book was such that John forgot to give it to Mary); sentence (55b), on the other hand, has the interpretation that John

was supposed not to give any books to Mary but he forgot to do so (i.e., he gave at least one book to Mary).

- (55) a. O João não esqueceu de dar *nenhum livro_F* pra Maria.
 the John not forgot of give no book to-the Mary
 ‘John did not forget to give *any book_F* to Mary.’
nenhum livro > *esqueceu*
- b. O João esqueceu de não dar *nenhum livro_F* pra Maria.
 the John forgot of not give no book to-the Mary
 ‘John forgot not to give *any book_F* to Mary.’
esqueceu > *nenhum livro*

Regarding (55a), it is easy to observe that the fact that *nenhum livro* ‘no book’ takes scope over *esqueceu* ‘forgot’ can be derived either by a structure meant to comply with the Neg Criterion or by a structure meant to comply with the Focus Criterion, as both options would place *nenhum livro* above *esqueceu* at LF. In (55b), however, if *nenhum livro* underwent covert focus movement, as represented in (56), it would necessarily take scope over *esqueceu*, contrary to fact, and furthermore violate the Neg Criterion. Again, if we choose to preserve the Neg Criterion and dispense with the Focus Criterion, the correct interpretation is straightforwardly derived, as represented in (57).

- (56) LF: [_{FocP} *nenhum livro_F* [_{TP} o João esqueceu de [_{NegP} ~~tf não~~ [_{TP} dar _{tf} pra Maria]]]]

cf. (55b)

(57) LF: [_{TP} o João esqueceu de [_{NegP} *nenhum livro*_F ~~não~~ [_{TP} dar t_F pra Maria]]]

cf. (55b)

While (57) gives us the correct interpretation of (55b) without resorting to the Focus Criterion, one could still try to salvage it by appealing to a focus position in the periphery of the embedded clause, as in (58), which would correctly place *nenhum livro* under the scope of *esqueceu*. However, there is reason to believe that such focus position is in fact not available.

(58) LF: [_{TP} o João esqueceu de [_{FocP} *nenhum livro*_F [_{NegP} t_F ~~não~~ [_{TP} dar t_F pra Maria]]]]

cf. (55b)

Note that the embedded sentences in (55) above can be analyzed as a type of restructuring clause that may lack a CP field altogether. As Wurmbrand (2014, 2015) argues, the possibility of negative concord across the clause boundary (as in (55a) above) indicates that the infinitival clause is of reduced size. Thus, expectedly, overt focus movement to the left periphery of the matrix clause is possible, whereas overt focus movement to the periphery of the embedded clause is ruled out, as the contrast in (59) shows. Needless to say, the structure in (58) would still violate the Neg Criterion, which I maintain we need to keep in order to account for the contrasts discussed in this section.

- (59) a. *Só dois livros_F o João esqueceu de dar pra Maria.*
 only two books the John forgot of to.give to-the Mary
- b. **O João esqueceu de só dois livros_F dar pra Maria.*
 the John forgot of only two books to.give to-the Mary
 ‘John forgot to give *only two books_F* to Mary.’

Before ending this discussion, I will now argue that the postulation of low focus movement (i.e., focus movement to a position in the periphery of vP, as discussed in Section 3) cannot be invoked in order to try to salvage the Focus Criterion in the event of a conflict with the requirements of the Neg Criterion. Let us then take another look at sentence (52B), repeated below in (60).

- (60) *Duas crianças não ganharam nenhum brinquedo_F.* = (52B)
 two children not received no toy
 ‘Two children did not receive *any toy_F*.’

If *nenhum brinquedo* ‘no toy’ moved (either overtly or covertly) to Spec,FocP in the periphery of vP in order to comply with the Focus Criterion, as in (61a), it would become frozen in place (cf. the criterial freezing) and thus unable to also satisfy the Neg Criterion at LF in order to license its negative quantificational force; in a structure where *nenhum*

brinquedo moves to Spec,NegP, as in (61b), it would then violate the Focus Criterion, which cannot be satisfied *en passant*.

- (61) a. LF: *_{[TP duas crianças [NegP Ø não [TP ganharam [FocP *nenhum brinquedo*_F [VP [VP t_F]]]]]]]} cf. (61)
- b. LF: *_{[TP duas crianças [NegP *nenhum brinquedo*_F não [TP ganharam [FocP t_F [VP [VP t_F]]]]]]]} cf. (61)

The same problems are replicated with respect to the sentences in (55) above, whose structures with the postulation of low focus movement are given in (62) below. In the LF representation of (55a) given in (62a), *nenhum livro* ‘no book’ would not be able to comply with the Neg Criterion and would crucially take scope under the matrix modal verb *esqueceu* ‘forgot’, contrary to fact. Similarly, the LF structure in (62b), though giving rise to the correct scope relation, would violate the Neg Criterion. Needless to say, movement of *nenhum livro* from Spec,FocP to Spec,NegP in both structures would violate the Focus Criterion.

- (62) a. LF: *_{[TP O João [NegP Ø não [TP esqueceu de [TP dar [FocP *nenhum livro*_F [VP [VP t_F pra Maria]]]]]]]]} cf. (55a)
- b. LF: *_{[TP O João esqueceu de [NegP Ø não [TP dar [FocP *nenhum livro*_F [VP [VP t_F pra Maria]]]]]]} cf. (55b)

To conclude, when a conflict arises between the Neg Criterion and the Focus Criterion from the focalization of a negative concord item, the data discussed in this section indicate that the Neg Criterion is operative in Brazilian Portuguese, whereas the Focus Criterion is not. Dispensing with the latter and preserving the former gives us the right results with respect to the scope possibilities attested in sentences involving negative concord, which provides evidence against the postulation of covert focus movement in this language.⁹ In the next section, this claim will be corroborated by data from contrastive topicalization.

4.2. *Contrastive topicalization*

I will now argue that in sentences where a contrastive topic and a focus co-occur in Brazilian Portuguese, the assumption of covert focus movement to the left periphery may lead to illicit information structure configurations. That problem can be circumvented if we allow for foci to be licensed in situ.

While traditional aboutness topics (in the sense of Reinhart 1981; see also Bianchi & Frascarelli 2010) are standardly assumed to be interpreted relatively to a comment in a bipartite articulation of the clause as topic-comment, as in (63), contrastive topics are

⁹ Evidently, focalized elements may still undergo other types of covert movement, such as the movement of negative concord items to Spec,NegP and quantifier raising. With that in mind, the expression “focalization in situ” should be interpreted with a grain of salt, as a descriptive expression intended to mean that no *focus-driven* step of movement is involved at LF and that the element surfaces in its canonical position.

instead argued to be interpreted in association with a focus (see e.g. Büring 2003, 2016; Wagner 2012).

(63) [This book]_{TOPIC}, [I will give it to Mary]_{COMMENT}.

As Büring (2003) discusses, a question like *Who ate what?* in (64A) can be interpreted as asking (i) for each person what that person ate or (ii) for each food who ate that food. The answer in (64B1), where *Fred* is pronounced as a contrastive topic and *the beans* as a focus, addresses the first interpretation of the question (i.e., the people function as independent variables), whereas the answer in (64B2), where *Fred* is pronounced as a focus and *the beans* as a contrastive topic, addresses the second interpretation of the question (i.e., the foods function as independent variables).

(64) A: Who ate what?

B1: Well, *Fred*_{CT} ate *the beans*_F.

B2: Well, *Fred*_F ate *the beans*_{CT}.

(Büring 2003: 511–512, adapted)

The distribution of contrastive topics relatively to their associated foci is a complex matter and subject to much ongoing debate (see e.g. Neeleman & van de Koot 2008, 2010, Lacerda 2020, and references above), which is beyond the scope of this paper. Still, two distributional restrictions observed in Brazilian Portuguese by Lacerda (2020) can be used

to our advantage to assess whether foci undergo covert movement in this language. I will argue that the postulation of covert focus movement fails to capture both restrictions and incorrectly allows the creation of illicit information structure configurations.

The first restriction is that contrastive topics must precede the focus in order to newly introduce (a relevant set of) contrastive alternatives in the discourse. More precisely, if the referent denoted by the contrastive topic is not known to be part of a relevant set of contrastive alternatives in the conversational context, the contrastive topic is only licit if it precedes the focus, as we will see below.¹⁰

Let us compare the dialogues in (65) and (66), where both answers are given in the canonical S–V–DO–IO order. Assume that Speaker A in (65) asks a question just about Barriers and no other books. In this case, Speaker B can reply as in (65B), with *o Barriers* marked as contrastive topic, raising the implicature that there are other books that John might have given to other people.

(65) A: Pra quem o João deu o Barriers?

‘Who did John give Barriers to?’

B: Bem, ele deu o *Barriers*_{CT} pra *Maria*_F.

¹⁰ Arguably, since the alternatives in the contrastive topic set constrain the alternatives in the focus set (i.e., the contrastive topic set contains independent variables, whereas the focus set contains dependent variables), the contrastive topic set must be computed prior to the focus set; in a nutshell, the contrastive topic must have wide scope over the focus in order to newly introduce (a relevant set of) contrastive alternatives in the discourse (see Wagner 2012 for relevant discussion).

well he gave the Barriers to-the Mary

‘Well, he gave *Barriers*_{CT} to *Mary*_F.’

Likewise, assume that in (66) Speaker A asks a question just about Mary and no other people. Now, Speaker B cannot (pragmatically felicitously) reply as in (66B), with *pra Maria* marked as a contrastive topic; that is, even though *pra Maria* is marked as a contrastive topic, the reply fails to introduce (a relevant set of) contrastive alternatives and raise the implicature that there are other people to whom John might have given books.

(66) A: Que livro o João deu pra Maria?

‘Which book did John give to Mary?’

B: #Bem, ele deu o *Barriers*_F pra *Maria*_{CT}.

well he gave the Barriers to-the Mary

‘Well, he gave *Barriers*_F to *Mary*_{CT}.’

In accordance with the above restriction, the sentence becomes acceptable with the relevant interpretation if the contrastive topic *pra Maria* ‘to Mary’ is dislocated to a position before the focus, as in (67B), or if Mary’s membership in a relevant set of contrastive alternatives is established in the discourse, as in (68B).

(67) A: Que livro o João deu pra Maria?

‘Which book did John give to Mary?’

B: Bem, *pra Maria_{CT}* ele deu *o Barriers_F*.

well to-the Mary he gave the Barriers

‘Well, *to Mary_{CT}* he gave *Barriers_F*.’

(68) A: Que livro o João deu pra Maria e (que livro ele deu) pro Pedro?

‘Which book did John give to Mary and (which book did he give) to Peter?’

B: Bem, ele deu *o Barriers_F* *pra Maria_{CT}*.

well he gave the Barriers to-the Mary

‘Well, he gave *Barriers_F* *to Mary_{CT}*.’

Taking advantage of the contrasts in (65)–(68) with respect to the restriction described above, let us now probe into the possibility of covert focus movement by examining the contrasts in (69) below. If the focus *só uma vez* ‘only once’ moved covertly in the grammatical sentence (69B1), it would create an LF structure analogous to sentence (69B2), which is however ruled out under the relevant information structure (with the direct object *o Barriers* interpreted and intonationally accented as a contrastive topic). Given that *Barriers* was not previously known (by Speaker A) to be a (contrastive) member of a relevant set of books, marking it as a contrastive topic in (69B2) is illicit, as it follows the

focus, and the reply fails to raise the implicature that there are other books that John read. Expectedly, sentence (69B3) is acceptable (cf. (67B) above).¹¹

(69) A: Quantas vezes o João leu o Barriers?

‘How many times did John read Barriers?’

B1: Bem, ele leu o *Barriers*_{CT} só uma vez_F.

well he read the Barriers only one time

B2: #Bem, só uma vez_F ele leu o *Barriers*_{CT} t_F.

well only one time he read the Barriers

B3: Bem, o *Barriers*_{CT}, só uma vez_F ele leu t_{CT} t_F.

well the Barriers only one time he read

‘Well, he read *Barriers*_{CT} only once_F.’

Suppose for the sake of the argument that, in addition to covert focus movement, we could also allow covert topic movement in (69), in order to try to salvage the necessary topic-focus order at LF. Sentence (69B1) would then have an LF structure analogous to sentence (69B3), which is well-formed, as the contrastive topic precedes the focus, a seemingly welcome result. However, resorting to covert topic movement in addition to focus movement would incorrectly rule in the ungrammatical sentence (69B2), which

¹¹ Neeleman & van de Koot (2008) discuss similar restrictions in Dutch scrambling and propose the following generalization: “while a topic can move out of a constituent containing a focus [...], a focus cannot move out of a constituent containing a topic [...]” (p. 146).

would then be indistinguishable from the grammatical (69B1) and (69B3) in the relevant respects at LF. We see then that neither the postulation of covert focus movement alone nor the postulation of covert topic and focus movement combined can account for the paradigm in (69), which is in turn accounted for if covert focus movement does not take place.

That focalization in situ cannot involve covert focus movement in Brazilian Portuguese is further corroborated by a second restriction, namely that contrastive topics cannot be dislocated to a position under the focus, which now gives rise to ungrammaticality and not just pragmatic infelicity. Let us first observe the scenario in (70), where the answers in (70B1) and (70B2) associate the contrastive topic *do Chomsky* ‘by Chomsky’ with the postverbal focus *só dois livros* ‘only two books’ (the question under discussion thus being for each author how many books by that author Speaker B read). Observe that the contrastive topic can be dislocated either to the left periphery of the sentence, as in (70B1), or to the postverbal middle field, as in (70B2); in both cases, the dislocated contrastive topic precedes the focus.

(70) A: Quantos livros você leu pro curso de sintaxe gerativa?

‘How many books did you read for the generative syntax class?’

B1: Olha, *do Chomsky*_{CT}, eu li *só dois livros*_F.

look of-the Chomsky I read only two books

B2: Olha, eu li, *do Chomsky*_{CT}, *só dois livros*_F.

look I read of-the Chomsky only two books

‘Look, I read *only two books_{sf} by Chomsky_{CT}*.’

Let us now look at (71) below. In contrast to (70B1) and (70B2) above, the focus associated with the contrastive topic in (71B1) and (71B2) is the preverbal subject *eu* ‘I’. Now, the topic *do Chomsky* ‘by Chomsky’ can be dislocated to the left periphery, as in (71B1), but not to the postverbal middle field, as in (71B2), which is utterly unacceptable (even if Chomsky is already known by Speaker A to be a part of the relevant set of authors; cf. the first restriction discussed above), in a sharp contrast with (70B2) above.

(71) A: Quem resenhou os livros pro curso de sintaxe gerativa?

‘Who reviewed the books for the generative syntax class?’

B1: Olha, *do Chomsky_{CT}*, *eu_F* resenhei os livros.

look of-the Chomsky I reviewed the books.

B2: *Olha, *eu_F* resenhei, *do Chomsky_{CT}*, os livros.

look I reviewed of-the Chomsky the books.

‘*I_F* reviewed all the books *by Chomsky_{CT}*.’

As the examples above show, movement to either the periphery of CP or the periphery of vP may serve to reposition a contrastive topic with respect to the focus in Brazilian Portuguese — which is successful in (70B1), (70B2), and (71B1), but not in (71B2).¹²

¹² See Lacerda (2020, 2021) for extensive discussion of middle-field topicalization in Brazilian Portuguese and arguments that it does not involve a dedicated topic projection TopP, but rather adjunction to the topmost

Taking advantage of the two possible landing sites for a dislocated contrastive topic, let us now assess the possibility of covert focus movement in these structures. Suppose that the focalized direct object *só dois livros* ‘only two books’ in (70B1) above moved covertly as in (72a) below. Assuming that this movement could target a position under the contrastive topic, (72a) would be a well-formed information structure configuration. Note, however, that if covert focus movement of *só dois livros* were obligatory in (70B1)/(72a), it would also need to take place in (70B2) above, where the contrastive topic is in the postverbal middle field; that operation would then create the structure in (72b), where the focus is in a position above the dislocated contrastive topic — which is an illicit configuration, as the ungrammaticality of (71B2) above shows.¹³

- (72) a. LF: [*do Chomsky*_{CT} [_{FocP} *só dois livros*_F [_{TP} eu li [_{vP} [_{vP} t_F]]]]] cf. (70B1)
 b. LF: *[[_{FocP} *só dois livros*_F [_{TP} eu li [*do Chomsky*_{CT} [_{vP} [_{vP} t_F]]]]] cf. (70B2)

As was the case in the discussion of (69) above, the attempted introduction of covert topic movement to try to salvage covert focus movement would create LF structures that

position in the (extended) vP domain. Unlike foci (see Section 3), movement of contrastive topics to the periphery of vP in Brazilian Portuguese may give rise to a licit information structure configuration.

¹³ Considering that movement of the focus to a position in the periphery of vP was already ruled out in Section 3, the LF structure in (i) below would also not be an option here.

- (i) LF: [_{TP} eu li [*do Chomsky*_{CT} [_{FocP} *só dois livros*_F [_{vP} [_{vP} t_F]]]]] cf. (70B2)

are indistinguishable in the relevant respects for all sentences in (70) and (71), as schematized in (73), thus incorrectly ruling in the unacceptable (71B2).¹⁴

(73) LF: [_{TopP} TOPICCT [_{FocP} FOCUSF [_{TP} ...]]]

In conclusion, the contrasts discussed in this section cannot be captured under the assumption that covert focus movement takes place in Brazilian Portuguese, but can be accounted for if foci in situ are allowed to be syntactically licensed and interpreted (*qua* informational objects) in sentence-internal positions at LF. Contrastive topicalization therefore provides further evidence that Rizzi's (1997) Focus Criterion is not operative in Brazilian Portuguese.

5. Final remarks

The well-known observation that the canonical order in Brazilian Portuguese allows for the focalization of any of the sub-constituents of the sentence strongly indicates that this is a true focus-in-situ language, which raises the question of how focalization is licensed at the Syntax-LF interface, especially in light of a prominent theory such as the cartographic

¹⁴ The putative covert topic movement to the left periphery should in principle be possible from the middle-field position in (70B2) and (71B2) if the topic is assumed to be adjoined to vP (see note 12), rather in Spec,TopP, where it would be subject to the criterial freezing (see Section 4.1). At any rate, I maintain that covert topic movement does not take place, as argued in the text.

approach to information structure (Rizzi 1997, 2004, 2006, 2010, 2015, 2018; Belletti 2001, 2004), which assumes that foci must move to the specifier of a focus-designated projection (FocP) in order to be licensed, in compliance with Rizzi's (1997) *Focus Criterion*. Once it is demonstrated that postverbal focalization in Brazilian Portuguese is done truly in situ in overt syntax (that is, without focus-driven movement to Spec,FocP), we naturally arrive at the question of whether focus movement to the left periphery must take place at LF.

The investigation of two independent empirical domains — namely, negative concord and contrastive topicalization — provided evidence that covert focus movement is empirically and theoretically inadequate in Brazilian Portuguese. As a consequence, Rizzi's (1997) Focus Criterion can be said not to be operative and should thus be dispensed with in the analysis of focalization in this language. If this is correct, overt focus fronting cannot be tied to the formal licensing of focus interpretation in the language, but should rather be tied to other semantic and/or pragmatic interpretive effects (which is independently argued to be the case by Lacerda 2020, 2024a).

The conclusions of this paper regarding Brazilian Portuguese are in line with what Costa (2000) argued for European Portuguese (modulo the fact that, given the time when his analysis was elaborated, he did not consider and rule out the possibility that postverbal foci in European Portuguese could be located in the periphery of vP). All in all, the two varieties of the language can be said to point to the direction that, rather than making reference to a specific position in (the left periphery of) the clause, as in the cartographic approach, the licensing of foci at LF is done in the style of Rooth (1985), according to

whom “focused phrases are, or at least can be, interpreted ‘in place’” (p. 15). In this well-known approach, focalization of a given element (say, through the presence of a focus feature) simply indicates the presence of alternatives in the interpretation of the focalized element; crucially, no formally-driven movement to a designated focus position is required. Informally put, foci can be interpreted as such “within” the presupposition and do not need to be “outside” it. If that line of reasoning is on the right track, we may after all say (tongue in cheek) that Brazilian Portuguese (at least when it comes to focalization) “wears LF on its sleeve”.

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